Case Study: Loan Processing & Loan Repayment Analytics (Medium Challenge)

Difficulty: (Medium)

Estimated Time: 90–120 Minutes

Domain: Banking / Financial Services

Business Scenario

ABC Bank receives hundreds of loan applications every day.

The Loan Processing Department wants to automate the loan approval process and analyze loan repayment performance.

The bank maintains the following datasets:

- customers.csv
- loan_applications.csv
- loan_payments.csv

The management needs insights into:

- Eligible and rejected loan applications
- Customer repayment performance
- Defaulted loans
- Loan recovery percentage
- EMI payment status
- Branch-wise and city-wise loan statistics

Input Files

customers.csv

loan_applications.csv

loan_payments.csv

Part 1 – Read Data

Read all CSV files using Pandas.

customers.csv

loan_applications.csv

loan_payments.csv

Part 2 – Data Cleaning

Perform the following:

- Remove duplicate records
- Remove duplicate Loan IDs
- Check missing values
- Replace missing Salary with Median Salary
- Replace missing Credit Score with Mean Credit Score
- Convert ApplicationDate and PaymentDate to datetime
- Remove negative Loan Amounts
- Remove invalid EMI Amounts
- Remove future payment dates

Part 3 – Merge Datasets

Merge all datasets using

CustomerID

LoanID

Create a single dataframe containing

- Customer Name
- City
- Loan Type
- Loan Amount
- Credit Score
- Salary
- Loan Status
- EMI Amount
- Payment Status

Part 4 – Create New Columns

Monthly Income

Monthly Income = Salary / 12

Debt-to-Income Ratio

Loan Amount / Salary

EMI Due

EMI Due = EMI Amount - Amount Paid

Payment Completion %

Amount Paid / EMI Amount × 100

Part 5 – NumPy Tasks

Using NumPy calculate

- Average Loan Amount
- Median Loan Amount
- Maximum Loan Amount
- Minimum Loan Amount
- Standard Deviation
- Variance
- 25th Percentile Loan Amount
- 75th Percentile Loan Amount

Part 6 – Pandas Analysis

Find

- Top 10 highest loan customers
- Top 10 customers by salary
- Customers with Credit Score below 650
- Customers with Loan Amount greater than ₹20 Lakhs
- Loans with Pending Payments
- Fully Paid Loans

Part 7 – GroupBy

Group by City

Find

- Number of Customers
- Average Salary
- Total Loan Amount

Group by Loan Type

Find

- Number of Loans
- Average Loan Amount
- Total Loan Amount

Group by Loan Status

Find

- Approved Loans
- Pending Loans
- Rejected Loans

Group by Payment Status

Find

- Paid
- Partial
- Pending

Count and Total Amount Paid

Part 8 – Business Rules

Flag the following loans:

- Loan Amount greater than ₹30 Lakhs
- Credit Score below 650
- Salary less than ₹30,000
- Debt-to-Income Ratio greater than 5
- EMI Due greater than ₹10,000
- Payment Status = Pending
- Loan Status = Rejected

Part 9 – Finance Metrics

Calculate

Total Loan Portfolio

Sum of Loan Amount

Total Amount Collected

Sum of Amount Paid

Outstanding Amount

Loan Amount - Amount Paid

Loan Recovery %

Amount Collected / Total Loan Amount × 100

Default %

Pending Loans / Total Loans × 100

Average EMI

Average EMI Amount

Average Credit Score

Average Credit Score

Part 10 – Export Reports

Generate

LoanSummary.xlsx

CustomerLoanReport.xlsx

PendingPayments.csv

Expected Outputs

Display

Top 10 Loan Customers

Customers with Low Credit Score

Pending Loan Payments

City-wise Loan Summary

Loan Type Summary

Loan Recovery Report